Solar Energy Systems

A solar photovoltaic panel converts sunlight into direct-current electricity. Panel output depends on sunlight intensity, panel temperature, orientation, and shading. Even partial shade can reduce output because cells in a panel are electrically connected. An inverter converts the panels' direct-current electricity into alternating current for building appliances or the electric grid. A battery stores excess electricity for use later, such as after sunset or during a short grid outage. A typical rooftop system connects panels, an inverter, safety equipment, and optionally a battery. Regular inspection helps identify dirt, damage, or wiring problems that can lower production.